

All-period resonant counts for nonlinear Hénon–Frobenius maps

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Abstract

Let $q = p^e$, let $K = \overline{\mathbb{F}}_q$, and consider $H(x, y) = (y, y^q + g(y) - ax)$ with $a \in \mathbb{F}_q^*$ and $g \in \mathbb{F}_q[y]$ of degree $2 \leq m < q$, where $p \nmid m$. For every positive integer n , the coincidence scheme $H^n(P) = (x^{q^n}, y^{q^n})$ is finite and reduced, with geometric point count

$$N_n = \frac{(m-1)q^{2n} + (q-m)q^{2n-p^{v_p(n)}}}{q-1}.$$

The count is independent of all lower coefficients of g and of its nonzero leading coefficient. The proof computes the unique leading term of every iterate of the difference of two commuting linear pullback operators; a characteristic- p factorization then resolves the cancellation at every period, including all levels of the p -divisibility tower. Coprime leading monomials and a nonvanishing Jacobian convert this degree calculation into an ordinary point count. These are precisely the fixed counts of the single map $S = H^{-1} \circ \Phi_q$. Its dynamical zeta function has an explicit convergent product, and every positive integer power has $|t| = q^{-2}$ as a natural boundary for meromorphic continuation. We distinguish this nonlinear count from established algebraic-group mechanisms, give a direct vector-group conjugacy obstruction, and exhibit a counterexample when the coprimality hypothesis is removed.

Keywords: Hénon automorphism; positive characteristic; Frobenius; resonant fixed points; dynamical zeta function; natural boundary.

1 Introduction

In positive characteristic, equality of the dominant degrees in a Frobenius coincidence problem can conceal a substantial cancellation. For the Hénon automorphism

$$H(x, y) = (y, y^q + g(y) - ax) \tag{1.1}$$

over $\mathbb{F}_q$, both H^n and the n th iterate of q -power Frobenius have degree q^n . In their second coordinates the dominant monomials cancel. The remaining degree, rather than the original degree, controls the affine intersection. A calculation at periods prime to the characteristic does not settle this problem: additional cancellations appear when the period acquires successive factors of p .

We compute the remaining degree for all periods and all coefficients in the following family. Throughout the paper,

$$q = p^e, \quad K = \overline{\mathbb{F}}_q, \quad a \in \mathbb{F}_q^*, \quad g \in \mathbb{F}_q[y], \quad 2 \leq m := \deg g < q, \quad p \nmid m. \tag{1.2}$$

The leading coefficient of g is denoted by $b \in \mathbb{F}_q^*$; no condition is imposed on its other coefficients. Set

$$\Phi(x, y) = (x^q, y^q), \quad S = H^{-1} \circ \Phi, \quad E_n = \{H^n(P) = \Phi^n(P)\} \subset \mathbb{A}_K^2.$$

Here E_n denotes the closed equalizer scheme. Its geometric points are the fixed points of S^n , as will also hold scheme-theoretically.

Theorem 1.1 (All-period resonant count). *Assume (1.2). For every $n \geq 1$, the scheme E_n is finite and reduced. Write $r = p^{v_p(n)}$ and $s = n/r$. Then*

$$N_n := \#E_n(K) = \frac{(m-1)q^{2n} + (q-m)q^{2n-r}}{q-1}. \quad (1.3)$$

More precisely, the two literal affine equations of E_n have respective unique highest homogeneous terms

$$-x^{q^n}, \quad sb^r \overline{m}^{r-1} y^{d_n}, \quad d_n = q^{n-r} \frac{(m-1)q^r + (q-m)}{q-1}, \quad (1.4)$$

where $\overline{m} \in \mathbb{F}_p^*$ is the residue of m . Their actual degrees are q^n and d_n .

The expression for d_n is an integer by the recurrence proved below; its displayed denominator is an integer denominator, not division in the coefficient field. In contrast, the coefficient s in (1.4) is read in $\mathbb{F}_p^*$ and survives because $p \nmid s$. This distinction separates the arithmetic of degrees from the characteristic- p cancellation of coefficients.

The exact count also determines the analytic behavior of the associated Artin–Mazur dynamical zeta germ

$$Z_S(t) = \exp \left(\sum_{n \geq 1} N_n \frac{t^n}{n} \right). \quad (1.5)$$

Corollary 1.2 (Product and natural boundary). *Put $B = (q-m)/(q-1)$ and $u = q^2 t$. With each logarithm normalized to vanish at zero,*

$$Z_S(u/q^2) = (1-u)^{-m/q} \prod_{k \geq 1} (1-u^{p^k})^{e_k}, \quad e_k = \frac{B}{p^k} (q^{-p^{k-1}} - q^{-p^k}) > 0. \quad (1.6)$$

The product converges locally uniformly on $|u| < 1$. For every fixed integer $M \geq 1$, the circle $|t| = q^{-2}$ is a natural boundary for meromorphic continuation of $Z_S(t)^M$. In particular, Z_S is neither rational nor algebraic over $\mathbb{C}(t)$.

The algebraic point of the paper is the all-period cancellation formula (1.4), not a new general principle relating distorted counts to natural boundaries. That principle has important antecedents. Byszewski, Cornelissen, and Houben study endomorphisms of algebraic groups with finite fixed sets at every iterate, develop finite-adelically distorted counting sequences, and prove corresponding zeta dichotomies [3, Theorems A and C, and §5.2]. Their vector-group theorem applies to additive group endomorphisms, whereas (1.1) yields a genuinely nonlinear affine-plane problem. Proposition 6.1 gives a fixed-cardinality obstruction to direct vector-group conjugacy in any dimension. It does not exclude every possible group quotient presentation.

One-dimensional polynomial dynamics also has well-established positive-characteristic zeta phenomena. In the version of Bridy’s paper specified in the bibliography, Theorem 1 gives rationality when the derivative of a polynomial on $\mathbb{A}^1$ is zero, and Theorem 2 gives transcendental examples [1]. These statements concern one variable, so neither determines the

counts of the two-dimensional map S . We make no claim to the first positive-characteristic transcendental dynamical zeta function.

Quotients require a separate distinction. Theorems A and B of the specified preprint of [2] concern dynamically affine maps on $\mathbb{P}^1$ and specified Kummer varieties; that work also supplies a higher-dimensional framework under additional hypotheses. We neither construct such a quotient for S nor verify those hypotheses for it. The present proof instead works directly in the polynomial ring. The source comparison identifies the nonlinear leading-degree calculation as the specific contribution considered here; it is a bounded comparison, not a certificate of global priority.

Section 2 identifies the single dynamical system and its pullbacks. Section 3 resolves the leading terms at all periods. Section 4 proves finiteness, length, and reducedness. Section 5 derives the product and proves its boundary statement directly. Section 6 records precise scope and bounded exact checks.

2 One dynamical system and its pullbacks

The inverse of (1.1) is the polynomial map

$$H^{-1}(x, y) = (a^{-1}(x^q + g(x) - y), x). \quad (2.1)$$

All coefficients lie in $\mathbb{F}_q$, so H commutes with Φ . It follows that

$$S^n = H^{-n}\Phi^n, \quad S(x, y) = (a^{-1}(x^{q^2} + g(x^q) - y^q), x^q). \quad (2.2)$$

Postcomposing both morphisms in an equalizer with an isomorphism preserves that equalizer. Applying H^n to the pair (S^n, id) therefore gives an equality of closed subschemes

$$\text{Fix}(S^n) = E_n. \quad (2.3)$$

This is not only a bijection of geometric point sets.

The map S is finite and radicial, since it is the composition of a polynomial automorphism with q -power Frobenius. Its function-field degree is q^2 , entirely inseparable. The degree here is not the cardinality of a geometric fiber: a radicial map has at most one point in such a fiber. Neither that degree nor radiciality by itself computes the fixed-point count.

Let $R = \mathbb{F}_q[x, y]$. We use the commuting $\mathbb{F}_q$ -linear ring endomorphisms

$$T(f) = f \circ H, \quad U(f) = f \circ \Phi = f(x^q, y^q) = f^q, \quad \delta = T - U. \quad (2.4)$$

The equality $U(f) = f^q$ holds because every coefficient of f is fixed by q -power Frobenius. Arbitrary elements of $K[x, y]$ do not satisfy that identity, which is why the operator calculation is performed in R before extending scalars for the zero scheme.

Although T and U are ring endomorphisms, their difference δ is in general only a linear operator. It is neither a ring endomorphism nor a derivation. Its iterates below mean composition as a linear operator, and every binomial identity takes place in the algebra of commuting linear operators on R . In particular,

$$\delta(y) = g(y) - ax, \quad \delta(x) = y - x^q. \quad (2.5)$$

There is no binomial identity asserted for a difference of nonlinear maps of the affine plane.

3 Leading terms through the entire divisibility tower

All degrees in this section are total polynomial degrees. Saying that P has unique highest homogeneous term cy^D means that $P - cy^D$ has degree at most $D - 1$. This stronger assertion, not merely the value of $\deg P$, prevents hidden cancellations after substitution.

Lemma 3.1. *Let $P \in R$ have unique highest homogeneous term cy^D , where $c \neq 0$ and $p \nmid D$. Then $\delta(P)$ has unique highest homogeneous term*

$$c\overline{D}by^{q(D-1)+m}. \quad (3.1)$$

Here $\overline{D}$ is the residue of D in $\mathbb{F}_p$.

Proof. Write $P = cy^D + P_0$ with $\deg P_0 \leq D - 1$, and set $h = g(y) - ax$. The coordinate degrees of H and Φ are at most q , so both $T(P_0)$ and $U(P_0)$ have degree at most $q(D - 1)$. The contribution of the top monomial is

$$c((y^q + h)^D - y^{qD}) = c \sum_{j=1}^D \binom{D}{j} y^{q(D-j)} h^j.$$

Since $\deg h = m$ and its unique top term is by^m , the term with $j = 1$ has the nonzero unique top monomial in (3.1). For $j \geq 2$, the degree is at most

$$q(D - j) + jm = q(D - 1) + m - (j - 1)(q - m),$$

strictly smaller because $m < q$. The contribution from P_0 also has strictly smaller degree. Thus no other term can cancel the stated monomial. $\square$

Lemma 3.2. *For every $j \geq 1$, the unique highest homogeneous term of $\delta^j(y)$ is $c_j y^{D_j}$, where*

$$c_j = b^j \overline{m}^{j-1} \neq 0, \quad (3.2)$$

$$D_1 = m, \quad D_{j+1} = qD_j - (q - m), \quad (3.3)$$

$$D_j = \frac{(m - 1)q^j + (q - m)}{q - 1}. \quad (3.4)$$

In particular, $D_j \equiv m \pmod{p}$ and $0 < D_j < q^j$.

Proof. For $j = 1$ the assertion follows from (2.5), since $m \geq 2$. If the assertion holds at j , then $D_j \equiv m \pmod{p}$, so Lemma 3.1 applies. It gives

$$D_{j+1} = q(D_j - 1) + m, \quad c_{j+1} = c_j \overline{m} b.$$

The degree recurrence again gives $D_{j+1} \equiv m \pmod{p}$, closing the induction. The coefficient recurrence yields (3.2). Solving the degree recurrence gives

$$D_j = q^j - (q - m)(1 + q + \cdots + q^{j-1}),$$

equivalent to (3.4). Positivity and the strict upper bound follow either from this expression and (3.4), or by induction using $1 < m < q$. $\square$

The hypothesis $p \nmid m$ is used at every step, not only at the first period: it ensures that the first noncancelled binomial term always has nonzero coefficient. Its failure is treated explicitly in Example 6.2.

Proposition 3.3. *For $n = rs$, where $r = p^{v_p(n)}$ and $p \nmid s$, the polynomial $T^n(y) - U^n(y)$ has unique highest homogeneous term*

$$sc_r y^{q^{n-r} D_r}. \quad (3.5)$$

The polynomial $T^n(x) - U^n(x)$ has unique highest homogeneous term $-x^{q^n}$.

Proof. Commutativity of T and U and the characteristic- p binomial identity give

$$T^r - U^r = (T - U)^r = \delta^r.$$

The difference-of-powers factorization in the same commuting operator algebra then gives

$$T^n - U^n = \sum_{i=0}^{s-1} T^{ri} U^{r(s-1-i)} \delta^r. \quad (3.6)$$

For any polynomial in R with unique top term cy^D , applying either T or U gives unique top term cy^{qD} . For T , the monic y^q term in the second coordinate supplies it, while every lower term has substituted degree at most $q(D-1)$. For U this is immediate. Coefficients in $\mathbb{F}_q$ remain unchanged.

Lemma 3.2 therefore shows that every summand of (3.6), applied to y , has unique highest homogeneous term

$$c_r y^{q^{r(s-1)} D_r} = c_r y^{q^{n-r} D_r}.$$

The s such terms add to a nonzero coefficient, because $p \nmid s$. This proves (3.5), including $s = 1$.

Finally, $T^n(x) = T^{n-1}(y)$ has degree q^{n-1} and unique top term $y^{q^{n-1}}$, by the same substitution observation starting with y . Since $U^n(x) = x^{q^n}$ has strictly larger degree, the first equation has the asserted top term. $\square$

4 From actual degrees to geometric points

We now prove Theorem 1.1. Put $Q = q^n$ and $d = d_n = q^{n-r} D_r$, and write the literal fixed equations as

$$F_1 = T^n(x) - x^Q, \quad F_2 = T^n(y) - y^Q.$$

Proposition 3.3 gives unique highest homogeneous terms $-x^Q$ and $sc_r y^d$, respectively. In particular, their leading monomials for any graded monomial order are relatively prime.

4.1 The quotient length

The coprime-leading-monomial criterion and the standard-monomial basis are classical polynomial-ideal tools; see [4]. Their application here can be made explicit. Normalize the polynomials to be monic and write

$$f = x^Q + f_0, \quad h = y^d + h_0,$$

where $\deg f_0 < Q$ and $\deg h_0 < d$. Their sole S -polynomial has the representation

$$y^d f - x^Q h = f_0 h - h_0 f.$$

Both products on the right have leading monomial strictly smaller than $x^Q y^d$. This is the standard representation in Buchberger's criterion, so f, h form a Gröbner basis. The monomials not in the initial ideal (x^Q, y^d) are exactly

$$x^i y^j, \quad 0 \leq i < Q, \quad 0 \leq j < d. \quad (4.1)$$

Their residue classes form a K -basis of the quotient. Thus E_n is finite and

$$\dim_K K[x, y]/(F_1, F_2) = Qd. \quad (4.2)$$

This calculation also shows why there is no unaccounted correction at infinity. Homogenizing each equation to its actual degree yields the forms $-x^Q$ and $sc_r y^d$ on the line at infinity. They cannot vanish at the same point of that projective line: simultaneous vanishing would require both homogeneous coordinates to be zero. Plane Bézout consequently gives the same length Qd . Homogenizing the second equation as if it still had degree Q would retain a spurious infinity contribution; resolving the resonant cancellation must precede the count.

4.2 Reducedness and the first period

The derivatives of x^Q and y^Q vanish in characteristic p . Moreover,

$$DH = \begin{pmatrix} 0 & 1 \\ -a & g'(y) \end{pmatrix}, \quad \det DH = a.$$

The chain rule therefore gives

$$\det \frac{\partial(F_1, F_2)}{\partial(x, y)} = \det D(H^n) = a^n \neq 0. \quad (4.3)$$

Each point of the finite zero scheme has zero-dimensional regular local ring. Over the algebraically closed field K , that ring is K . The scheme is therefore reduced, and its length is its number of geometric points. Substituting (3.4) into (4.2) gives

$$N_n = Qq^{n-r} D_r = q^{2n-r} \frac{(m-1)q^r + (q-m)}{q-1},$$

which is (1.3). This completes the proof of Theorem 1.1.

At $n = 1$, the equations are simply $y = x^q$ and $g(y) - ax = 0$. Eliminating y gives $g(x^q) - ax = 0$, of degree qm and derivative $-a$. Thus $N_1 = qm$ with simple roots, in agreement with the full argument. No exceptional convention is required at the first period. In particular, neither reducedness nor the passage from ideal length to point count has been inferred from finite computational agreement.

5 The zeta product and all positive powers

This section proves Corollary 1.2 directly from the count. No algebraic-group structure or abstract zeta dichotomy is required for the proof. Set

$$A = \frac{m-1}{q-1}, \quad B = \frac{q-m}{q-1}, \quad \mathcal{Z}(u) = Z_S(u/q^2).$$

Both A and B are positive and $A + B = 1$. Formula (1.3) becomes

$$\frac{N_n}{q^{2n}} = A + Bq^{-p^{v_p(n)}}. \quad (5.1)$$

Its right side lies between A and $A + B/q = m/q$, so $N_n^{1/n} \rightarrow q^2$. In particular, the logarithmic series defining $\mathcal{Z}$ converges absolutely on $|u| < 1$.

5.1 Divisibility decomposition and convergence

For every positive integer n the following sum is finite and telescopes:

$$q^{-p^{v_p(n)}} = q^{-1} + \sum_{k \geq 1} (q^{-p^k} - q^{-p^{k-1}}) \mathbf{1}_{p^k | n}. \quad (5.2)$$

For $|u| \leq \rho < 1$, the absolute sum needed to interchange n and k is bounded by

$$\sum_{k \geq 1} (q^{-p^{k-1}} - q^{-p^k}) \sum_{n \geq 1} \frac{\rho^n}{n} = q^{-1} [-\log(1 - \rho)] < \infty.$$

Using

$$\sum_{p^k | n} \frac{u^n}{n} = \frac{1}{p^k} \sum_{j \geq 1} \frac{u^{p^k j}}{j} = -\frac{1}{p^k} \log(1 - u^{p^k}),$$

we obtain

$$\log \mathcal{Z}(u) = -\frac{m}{q} \log(1 - u) + \sum_{k \geq 1} e_k \log(1 - u^{p^k}), \quad (5.3)$$

with the coefficients e_k in (1.6). Every logarithm denotes its analytic branch on the unit disk with value zero at zero. The positive coefficients e_k are summable. Also, on $|u| \leq \rho < 1$,

$$|\log(1 - u^{p^k})| \leq \sum_{j \geq 1} \frac{\rho^{p^k j}}{j} \leq -\log(1 - \rho).$$

Thus the logarithmic sum converges locally uniformly. Exponentiating gives the locally uniformly convergent product (1.6); it is analytic and nonzero on $|u| < 1$ and equals the original germ there.

5.2 Fractional radial orders

Let ξ be a primitive p^a -th root of unity, where $a \geq 1$. For $k < a$, $1 - \xi^{p^k} \neq 0$, and also $1 - \xi \neq 0$. The associated finitely many logarithmic moduli stay bounded along $u = \rho\xi$ as $\rho \uparrow 1$. For $k \geq a$, one has $\xi^{p^k} = 1$, and

$$0 \leq \frac{\log(1 - \rho^{p^k})}{\log(1 - \rho)} \leq 1, \quad \lim_{\rho \uparrow 1} \frac{\log(1 - \rho^{p^k})}{\log(1 - \rho)} = 1. \quad (5.4)$$

The inequality follows from $1 - \rho \leq 1 - \rho^{p^k} < 1$; the limit follows from $1 - \rho^{p^k} = (1 - \rho)(1 + \rho + \dots + \rho^{p^k-1})$. Taking real parts in (5.3), dividing by $\log(1 - \rho)$, and applying dominated convergence to the summable weights e_k yields the exact radial order

$$\lim_{\rho \uparrow 1} \frac{\log |\mathcal{Z}(\rho\xi)|}{\log(1 - \rho)} = \sigma_a := \sum_{k \geq a} e_k. \quad (5.5)$$

It satisfies

$$0 < \sigma_a \leq \frac{B}{p^a} \sum_{k \geq a} (q^{-p^{k-1}} - q^{-p^k}) = \frac{Bq^{-p^{a-1}}}{p^a} < 1, \quad \sigma_a \longrightarrow 0. \quad (5.6)$$

If a nonzero meromorphic function is defined in a neighborhood of ξ , it has the form $(u - \xi)^j h(u)$ there, where $j \in \mathbb{Z}$ and h is holomorphic and nonzero at ξ . Its radial logarithmic

order is j , since $|\rho\xi - \xi| = 1 - \rho$ and $\log|h(\rho\xi)|$ stays bounded. The strictly fractional order σ_a consequently excludes a meromorphic extension of $\mathcal{Z}$ to a neighborhood of ξ .

Now fix any integer $M \geq 1$. By (5.6), for every sufficiently large a one has $0 < M\sigma_a < 1$. The same argument excludes a meromorphic extension of $\mathcal{Z}^M$ at every primitive p^a -th root of unity for all such a . These roots are dense on the unit circle: among the grid numerators modulo p^a , excluding the multiples of p leaves gaps of at most two, so the largest angular gap tends to zero with a . An extension across any nonempty arc would give an extension at one of these forbidden points. Hence the unit circle is a natural boundary for meromorphic continuation of $\mathcal{Z}^M$.

Returning to $t = u/q^2$ proves the asserted circle for Z_S^M . Finally, an algebraic function over $\mathbb{C}(u)$ has only finitely many possible finite branch points or poles: away from the zeros of the leading coefficient and discriminant of a square-free defining polynomial, its local branches are holomorphic. It cannot have a natural boundary along the unit circle. Thus Z_S is not algebraic, and in particular is not rational. This proves Corollary 1.2.

6 Model boundaries and bounded exact checks

6.1 A direct additive-model obstruction

The first coordinate of S contains the nonzero term $a^{-1}bx^{mq}$, where mq is not a power of p . Thus S is not additive for the standard vector-group law on $\mathbb{A}^2$. That coordinate observation alone would not exclude a different coordinate system. The first-period count gives a stronger but still limited statement.

Proposition 6.1. *The point dynamical system $S: \mathbb{A}^2(K) \rightarrow \mathbb{A}^2(K)$ is not conjugate, even by a set-theoretic bijection, to an algebraic-group endomorphism of any vector group $\mathbb{G}_a^d(K)$ having finite fixed sets at every iterate.*

Proof. For a vector-group endomorphism τ , its fixed points form the subgroup $\ker(\tau - \text{id})(K)$ of $\mathbb{G}_a^d(K)$. If finite, this is a finite-dimensional vector space over $\mathbb{F}_p$, and therefore has cardinality a power of p . On the other hand, Theorem 1.1, or the first-period elimination in Section 4, gives $N_1 = qm$. Since $m > 1$ and $p \nmid m$, this is not a power of p . A conjugacy preserves fixed-set cardinalities, giving a contradiction. $\square$

This elementary obstruction distinguishes the family from the direct vector-group setting of [3, Theorem 5.2.5]. It says nothing about finite quotients, other algebraic groups, or a birational presentation restricted to a smaller domain. In particular, it is not a theorem excluding every dynamically affine presentation. Such a classification is not needed for the count proved here and is not supplied by it.

6.2 Why the coprimality hypothesis cannot be dropped

Example 6.2. Take $q = 8$, $a = 1$, $g(y) = y^6 + y^3 + 1$, and $n = 2$, so $p \mid m$. The geometric count is 2816, whereas extending (1.3) to these parameters would give 2944.

Proof. Work in characteristic two and put $h = y^6 + y^3 + 1 + x$. The commuting-operator identity still gives $T^2 - U^2 = \delta^2$, and direct expansion gives

$$\delta^2(y) = y^{32}h^2 + y^{16}h^4 + h^6 + y^{16}h + y^8h^2 + h^3 + y + x^8.$$

The unique top term is y^{44} : the first three displayed products have degrees 44, 40, 36, and the remaining degrees are smaller. The first equation has unique top term x^{64} . The quotient-length and Jacobian arguments of Section 4, here with determinant one, therefore give

$64 \cdot 44 = 2816$ reduced points. The unjustified extension of (1.3) gives $(5 \cdot 8^4 + 2 \cdot 8^2)/7 = 2944$, as claimed. $\square$

This is a nonlinear failure example, not an additive degeneration. It shows that deleting $p \nmid m$ is invalid; it does not classify every degree divisible by p .

6.3 Exact checks and their evidentiary role

The accompanying bounded calculation literally composes H and forms the two fixed polynomials. It does not construct them from the degree recurrence. Exact coefficient arithmetic then determines their degrees, highest homogeneous terms, and Jacobian. Table 1 records the five archived cases. These values are supplementary checks of the implementation and hypotheses, not the proof of a universally quantified statement.

For the genuine extension-field examples, we use $\mathbb{F}_4 = \mathbb{F}_2[\alpha]/(\alpha^2 + \alpha + 1)$ and

$$a = \alpha + 1, \quad g(y) = \alpha y^3 + (\alpha + 1)y^2 + y + \alpha.$$

Thus their leading and lower coefficients do not all lie in the prime field. In the archived encoding, the integers 2 and 3 mean α and $\alpha + 1$, respectively; arithmetic is not performed in the nonfield ring $\mathbb{Z}/4\mathbb{Z}$.

q	Coefficients and period	Degrees	Count
5	$a = 2, g = 2y^3 + 3y^2 + y + 4, n = 2$	(25, 15)	375
9	$a = 2, g = 2y^2 + y + 1, n = 3$	(729, 92)	67068
4	Genuine $\mathbb{F}_4$ map above, $n = 2$	(16, 11)	176
4	Same genuine $\mathbb{F}_4$ map, $n = 4$	(256, 171)	43776
8	Example 6.2, $n = 2$	(64, 44)	2816

Table 1: Bounded exact checks from the archived literal-composition run. The final row is outside the theorem’s hypotheses.

For the three prime-subfield coefficient cases, an additional F5B Gröbner calculation with SymPy computes the quotient length from a returned basis. That secondary method was not run for the two genuine $\mathbb{F}_4$ cases; no extension-field SymPy Gröbner check is claimed. The counts inferred from the coprime leading terms use the same elementary length principle as the proof and are not an independent proof of that principle. The checks include a prime-to- p period, a wild period over a nonprime base size, non-prime-field coefficients, a second p -power level, and an out-of-hypothesis control. They are not a large census or an empirical extrapolation to all periods.

6.4 What is, and is not, being counted

The clock is the positive integer n for the one fixed map S on $\mathbb{A}^2(K)$. Equivalently, both H and Φ are iterated n times in the coincidence condition. This is not the count of ordinary H -periodic points over a fixed finite field, and it is not the Hasse–Weil point count of a fixed variety under Frobenius alone. Accordingly, the natural boundary does not assert a failure of Hasse–Weil rationality.

All lower coefficients of g are permitted, but the monic y^q term, the coefficient field $\mathbb{F}_q$, $a \neq 0$, and $2 \leq m < q$ with $p \nmid m$ are structural hypotheses. The result is not a classification of all resonant Hénon maps, of independently varying two-clock resonances, or of the case $p \mid m$.

7 Conclusion and reproducibility

The all-period resonance in (1.1) is resolved by a coefficient-uniform leading-term recurrence in a polynomial ring. The characteristic- p factorization isolates the largest p -power dividing the period, and the remaining factor contributes a nonzero scalar. Together with the coprime leading monomials and Jacobian, this proves an exact geometric count for every period. The same count then gives one explicit dynamical zeta function and its natural boundary; the analytic consequence is not a separate counting theorem.

Several questions remain outside this result: a complete treatment of $p \mid \deg g$, nonmonic Frobenius-degree terms, independently varying clocks, and possible presentations through other groups or finite quotients. Nonrationality excludes a finite-dimensional characteristic-zero determinant formula for this exact zeta, since a finite product or quotient of polynomial determinants is rational. It does not exclude infinite-dimensional realizations. No target arithmetic Euler factors, zero correspondence, root numbers, automorphy statement, or Hilbert–Pólya realization are obtained.

Reproducibility. The accompanying package includes the full proof contract, a bounded primary-source audit, the exact-composition script `exact_checks.py`, its archived output `exact_results.json`, and `CHECK_RECEIPT.md`. The recorded successful run used Python 3.12.3 and SymPy 1.14.0. The receipt also records the initial JSON-serialization failure and its implementation repair; the mathematical formula was not changed to fit that failure. Preparing this manuscript did not rerun an older sealed numerical package. The universal proofs are all included in the present text, independently of executable availability.

Research and review disclosure. This manuscript was developed with AI assistance within the current research team, including internal non-author checks of the proof contract and source positioning. Such checks are not external human peer review and do not certify priority. The anonymous author block does not identify or imply an institution. Human authorship, contribution, funding, and competing-interest declarations have not been supplied; they must be resolved before any external submission.

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